output data is not satisfactory, modification is done to the algorithm and the training data by the supervisor. Then the agent is fed again with new data and the output is collected (as a feedback to the supervisor). In case the output is not satisfactory, more modifications are done and the cycle continues until the agent is stable and matured. Here, the Naive Bayes algorithm is used because it's best known for its ability to classify and predict data.

The Naive Bayes algorithm works on the following formula:

$$P(A|B)=\frac{P(B|A)\,P(A)}{P(B)}$$

where, A, B = events

P(A|B) = probability of A given B is true

P(B|A) = probability of B given A is true

P(A), P(B) = the independent probabilities of A and B

However, if you need to understand the above formula in simple English, then it is:

$$\text{Posterior} = \frac{\text{Prior} \times \text{Likelihood}}{\text{Evidence}}$$

The Posterior probability for supervised training would be the expected prediction data. To get that data, the Prior probability data is always updated. You can say that it's like history data. Likelihood is the data that you can assume from previous events, but Evidence is the data that is already known and correct.

Example. If the chance of getting attacked by a dangerous computer virus is 1 percent but the detection of any virus by cybersecurity software is 10 percent, and 90 percent of viruses are not harmful, then the probability of getting any harm from a dangerous virus would be:

$$P(\text{Harm}|\text{Virus}) = \frac{P(\text{Harm}) \times P(\text{Virus}|\text{Harm})}{P(\text{Virus})}$$

$$= \frac{1\% \times 90\%}{10\%} = 9\%$$

The algorithm is:

1. Separate the training data by class.

2. Summarise data sets by finding the mean and standard deviation using the formulae:

$\text{Mean} = \frac{\sum |X-\mu|}{N}$

where, x is the feature data and N is the number of feature data.

$\mu = (x_1 + x_2 + x_3 \ldots + x_n)/N$

Standard deviation $= \sqrt{\frac{1}{N}\sum_{i=1}^{N}}$ $(x_i - \mu)^2$, x_i is the featured data starting with the index value of i.

3. Summarise data by class.

4. Calculate the probability using the Gaussian Probability Density Function formula: $\frac{1}{\partial\sqrt{2\pi}}\, e^{-\frac{1}{2}\left(\frac{x-\mu}{\partial}\right)^2}$, where μ is the mean and ∂ is the standard deviation.

5. Predict class probability using the equation:

P(Class|dataset) = P(X|Class) × P (Class)

…which is a simplified version of the Naive Bayes formula.

Unsupervised learning

This learning approach requires no supervisor and the agent uses unlabelled data for input processing. The algorithm used in this learning approach helps the agent to find structure in its input, so that it can group or cluster data based on its patterns. Here, the input data is clustered by the agent accordingly, and can also discover hidden data. After clustering the data, the agent tries to find similarities between them and creates a relationship model. This learning requires a proper learning environment and the better the input data, the better is the learning. The algorithm used in this learning is called K-Means and the formula is:

$$J = \sum_{j=1}^{k} \sum_{i=1}^{n} \| x_i^{(j)} - c_j \|^2$$

Here,

J = objective function

k = number of clusters

n = number of cases

$x_i^{(j)}$ = case i

c_j = centroid for cluster j

$x_i^{(j)} - c_j$ = distance function

Clustering can also be done using a simple Euclidean distance formula:

$$\sqrt{(x_x - x_1)^2 + (x_y - y_1)^2}$$

Here, X_x and X_y are observed values and x_1 and y_1 are actual/centroid values.

Example. You can divide the following data set into two clusters:

Data ID	X	Y
1	1	2
2	3	4
3	5	6
4	7	8
5	9	10

First you find the centroids and create the two clusters (C1, C2) by taking the values of two Data IDs (1,2):

Clusters	X	Y	Centroid
C1	1	2	(1,2)
C2	3	4	(3,4)

Now you find in which cluster (C1 or C2) the next values of Data ID (3) will be included by using the Euclidean distance formula as follows:

$$C1 = \sqrt{(5-1)^2 + (6-2)^2}$$
$$= \sqrt{(4)^2 + (4)^2}$$
$$= \sqrt{32} = 5.65$$
$$C2 = \sqrt{(5-3)^2 + (6-4)^2}$$
$$= \sqrt{(2)^2 + (2)^2}$$
$$= \sqrt{8} = 2.28$$

Now, as the value of C2 is less than C1, the Data ID (3) will be clustered with C2.

Next, the new centroid values of C2 are:

$$C2 = ((X_{C2} + X_3)/2, (Y_{C2} + Y_3)/2)$$
$$= ((3+5)/2, (4+6)/2)$$
$$= 4,5$$

Similarly, the values for the next Data IDs can be found by using the new centroids of C2 and carrying on the clustering process.

The algorithm is:

1. Cluster data into k groups, where k is predefined.

2. Select random cluster centroids from the k points.

3. Allocate data to the nearest clusters according to the Euclidean distance calculations.

4. Calculate new cluster centroids for next data.

5. Repeat steps from 2 to 4 until no further clustering is possible.

Reinforcement learning

This type of learning works based on a reward/penalty policy and the agent is programmed to do certain predefined functions in an environment. After executing the functions, the agent gets feedback from the environment, either as a reward or penalty. This changes the state of the environment and the agent also gathers the state of the environment as an input. Depending on the state and reward values, the agent takes decisions. Depending on the rewards and state changes, the agent improves its quality of perfor-